

Efficient Human Activity Recognition Using Magnitude-Enhanced Statistical Features and Extra Trees Ensemble

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Abstract—Human Activity Recognition (HAR) is an important component in intelligent systems, with applications in healthcare, smart environments, and ubiquitous computing. Although deep learning models can achieve strong recognition performance, their computational cost and limited interpretability may restrict their use in resource-constrained scenarios. This paper presents a lightweight feature-based HAR framework using the WISDM dataset. The framework combines sliding-window segmentation with magnitude-enhanced statistical feature extraction and an Extra Trees classifier. The proposed feature representation is evaluated through baseline comparisons, ablation analysis, cross-dataset validation on UCI HAR, and explainability analysis using feature importance and SHAP. Experimental results show that the proposed approach achieves 97.57% accuracy on WISDM, outperforming the classical baselines evaluated under the same experimental setup, including Logistic Regression, SVM, and Random Forest, while remaining closely competitive with gradient boosting models. Comparisons with reported deep learning results are used as contextual references due to differences in preprocessing and evaluation protocols. Overall, the results suggest that magnitude-enhanced statistical features combined with ensemble learning can provide an efficient and interpretable option for practical HAR pipelines.

Index Terms—Human Activity Recognition, Feature Engineering, Ensemble Learning, Extra Trees Classifier, Time-Series Analysis, Sensor Data, Edge Computing

I. INTRODUCTION

Human Activity Recognition (HAR) is a core component of intelligent systems with applications in healthcare, assisted living, and IoT environments. Despite substantial progress, achieving high accuracy while maintaining efficiency and interpretability remains challenging, particularly in resource-constrained and real-time settings where sensor data is affected by noise, user variability, and device orientation changes.

However, further empirical evaluation is still needed to examine how magnitude-enhanced statistical representations perform within lightweight ensemble-based HAR pipelines, particularly when combined with ablation, explainability, and cross-dataset validation.

Many recent approaches rely on deep learning models such as CNNs, LSTMs, and GRUs, which can achieve strong performance but often introduce high computational cost and limited interpretability, making them challenging for resource-constrained deployment. In contrast, feature-based methods offer improved efficiency but remain underexplored and lack systematic validation at scale.

In this work, we propose a lightweight and interpretable HAR framework based on statistical feature engineering and ensemble learning. The approach integrates magnitude-enhanced, rotation-invariant features with an Extra Trees classifier to improve motion representation while maintaining low computational complexity. The randomized and parallel nature of Extra Trees further supports efficient inference in lightweight HAR settings.

Experimental results on the WISDM dataset show that the proposed framework achieves competitive performance compared with classical baselines and reported deep learning results, while maintaining low software-level computational overhead. Additional cross-dataset evaluation, ablation studies, and SHAP-based analysis are conducted to validate robustness and interpretability.

The main contribution of this work is the development and systematic evaluation of a lightweight HAR pipeline combining magnitude-enhanced statistical features with ensemble learning, rather than proposing a fundamentally new classifier. The framework evaluates Extra Trees against classical and ensemble baselines under consistent feature settings, while incorporating ablation analysis, cross-dataset validation on

UCI HAR, explainability using feature importance and SHAP, and computational efficiency measurements including training time, inference latency, and model size.

The remainder of this paper is organized as follows. Section II reviews related work in Human Activity Recognition. Section III presents the proposed methodology. Section IV describes the experimental setup, results, and evaluation analysis. Section V compares the proposed approach with gradient boosting methods. Section VI presents the explainability analysis using SHAP. Section VII discusses the study limitations. Finally, Section VIII concludes the paper and outlines future research directions.

II. RELATED WORK

A. Existing HAR Approaches

Human Activity Recognition (HAR) has been studied using both traditional machine learning and deep learning methods. Traditional approaches rely on handcrafted statistical or frequency-domain features with classifiers such as SVMs and decision trees, offering efficiency but limited robustness to sensor noise, user variability, and device orientation changes [1]–[3]. Deep learning models such as CNNs, LSTMs, and GRUs can learn features directly from raw sensor data, but often require higher computational resources and provide lower interpretability [2]–[4]. Lightweight alternatives, including knowledge distillation, model compression, and GADF-based methods, reduce complexity but may depend on auxiliary models or additional preprocessing stages [5]–[7].

B. Synthesis and Research Gap

Despite recent advances in HAR, balancing accuracy, efficiency, and interpretability remains challenging. Deep learning methods often require high computational cost and offer limited transparency, while lightweight approaches may lack sufficient validation or introduce pipeline complexity [8]. This motivates further investigation of magnitude-aware statistical features as a practical lightweight representation, as summarized in Table I.

TABLE I
COMPARATIVE ANALYSIS OF REPRESENTATIVE HAR METHODS

Study	Method	Strengths	Limitations
Fan et al. (2025) [1]	Residual BiGRU	Strong temporal modeling	High computational cost
Mekruksavanich et al. (2025) [2]	CNN + SE	Enhanced feature learning	High model complexity
Xiao et al. (2025) [5]	Knowledge distillation	Lightweight inference	Depends on teacher model
Lu et al. (2026) [7]	GADF + CNN	Rich signal representation	Additional preprocessing cost
Ameen et al. (2026) [8]	Robust CNN pooling	Improved noise robustness	Limited evaluation scope
Fatlawi et al. (2025) [4]	RL + LSTM	Adaptive decision fusion	Training instability
Proposed Work	Extra Trees	Magnitude-aware features and interpretability	Incremental feature-based contribution

C. Motivation for the Proposed Work

Existing HAR systems struggle to balance accuracy, efficiency, and interpretability. This work proposes a lightweight approach using magnitude-enhanced statistical features with an ensemble classifier to capture discriminative patterns efficiently. The goal is to investigate whether carefully designed statistical and magnitude-based features can provide competitive recognition performance with lower computational complexity and improved interpretability compared with more complex learning pipelines.

III. PROPOSED METHOD

A. Overview

The proposed framework follows a lightweight HAR pipeline, as shown in Fig. 1. Raw tri-axial accelerometer data are first preprocessed using filtering, interpolation, and normalization. The cleaned signals are then segmented using sliding windows, followed by magnitude computation to enhance motion representation. Statistical descriptors are extracted to form a compact magnitude-enhanced feature vector, which is used to train the Extra Trees classifier and compare it with baseline models. The framework further includes evaluation, explainability analysis, computational efficiency analysis, and cross-dataset validation to assess performance, interpretability, and robustness.

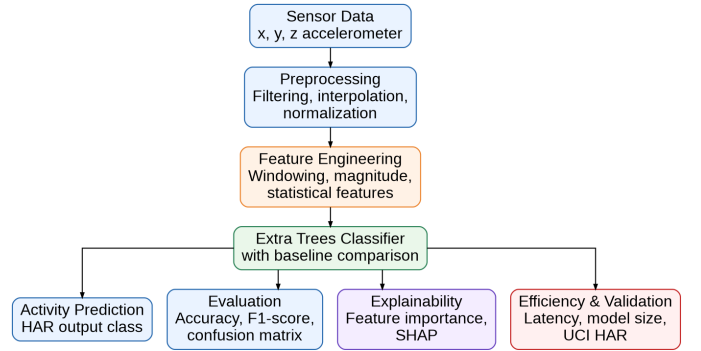


Fig. 1. Proposed HAR framework architecture

B. Data Preprocessing

Raw sensor signals are cleaned by removing invalid and missing values. To ensure orientation invariance, a magnitude feature is computed as:

$$m = \sqrt{x^2 + y^2 + z^2} \quad (1)$$

This provides a rotation-invariant representation of motion intensity.

C. Statistical Feature Extraction

Each window $X = \{x_1, x_2, \dots, x_n\}$ is converted into a feature vector using statistical descriptors such as mean, standard deviation, variance, minimum, maximum, median, and percentiles. These features capture both central tendency and variability without requiring sequential modeling.

D. Data Segmentation

A sliding window of size 100 with step 40 is used to balance temporal context and computational efficiency.

E. Feature Scaling and Encoding

Activity labels are encoded using LabelEncoder. Although tree-based models are scale-invariant, feature standardization is applied for fair comparison with SVM.

F. Baseline Models

The proposed method is compared with Logistic Regression, SVM, and Random Forest to cover linear, kernel-based, and ensemble learning approaches.

G. Extra Trees Classifier

The Extra Trees classifier is an ensemble of randomized decision trees that aggregates predictions via majority voting. Unlike Random Forest, it introduces additional randomness in split selection, which reduces variance and improves generalization. Its randomized and parallelizable structure can support efficient inference in lightweight HAR settings.

H. Class Imbalance Handling

The dataset shows mild class imbalance; therefore, no resampling is applied. Evaluation is performed using precision, recall, and F1-score to ensure balanced performance assessment across classes.

IV. EXPERIMENTAL SETUP

A. Dataset

Experiments are conducted on the WISDM dataset, a benchmark for smartphone-based Human Activity Recognition using tri-axial accelerometer signals. It contains six activities: Walking, Jogging, Upstairs, Downstairs, Sitting, and Standing. After preprocessing, 1,086,465 raw samples are segmented into 16,159 instances using a sliding window approach, where each instance is represented by statistical descriptors and a magnitude-based feature.

A stratified split is used to preserve class distribution, resulting in 9,715 training samples, 2,466 validation samples, and 3,978 testing samples.

B. Cross-Dataset Robustness Evaluation (UCI HAR)

To evaluate generalization, the proposed framework is tested on the UCI HAR dataset, which provides pre-extracted features, enabling evaluation under a different representation setting.

Adaptation experiments include feature scaling, feature reduction, and variation in ensemble size, as summarized in Table II.

TABLE II
ADAPTATION EXPERIMENTS ON UCI HAR DATASET

Experiment	Accuracy
No Scaling	0.9413
With Scaling	0.9406
Reduced Features	0.9192
Extra Trees (100 Trees)	0.9392
Extra Trees (300 Trees)	0.9413

The adaptation results indicate that scaling has negligible impact, while feature reduction reduces accuracy. As shown in Table III, the model also maintains strong macro-averaged performance on UCI HAR, supporting its robustness across different feature representations.

TABLE III
MACRO-AVERAGED PERFORMANCE ON UCI HAR DATASET

Precision	Recall	F1-score
0.9443	0.9396	0.9408

C. Evaluation Metrics

Performance is evaluated using Accuracy, Precision, Recall, and F1-score. Both macro and weighted averages are reported to handle class imbalance. Results are averaged over 5 independent runs to ensure stability.

D. Statistical Validation

A paired t-test is conducted to evaluate statistical significance between the proposed method and baseline models. The results show that the improvements of the Extra Trees classifier are statistically significant ($p < 0.05$). Additionally, a 95% confidence interval over 5 runs is reported to quantify result variability.

E. Implementation Details

The framework is implemented using Python and Scikit-learn. The evaluated models include Extra Trees, Random Forest, Logistic Regression, and SVM, trained under identical feature settings for fair comparison.

The Extra Trees classifier uses 600 trees with no maximum depth constraint and a minimum of 2 samples per split, with full parallelization enabled.

F. Computational Complexity Analysis

The model has training complexity $O(T \cdot n \log n)$ and inference complexity $O(T \log n)$, where T is the number of trees and n is the number of samples. Unlike deep learning methods, it avoids backpropagation, resulting in lower computational overhead.

Training takes approximately 4.16 seconds, while inference latency is 0.29 ms per sample, indicating low software-level inference overhead.

G. Memory Footprint and System Resource Analysis

The trained model size is approximately 0.34 MB. Peak memory usage (668 MB) is mainly due to preprocessing steps such as data loading and feature extraction. During inference, only the trained model and a single feature vector are required, ensuring minimal resource consumption.

H. Results and Discussion

To ensure statistical reliability, all experiments were repeated across multiple runs with different random seeds, showing low variance and confirming the stability of the proposed framework. Table IV summarizes the performance of all evaluated models on the WISDM dataset.

TABLE IV
PERFORMANCE COMPARISON OF MACHINE LEARNING MODELS

Model	Accuracy	Precision	Recall	F1-score
Logistic Regression	0.8820	0.8831	0.8820	0.8815
SVM (RBF)	0.9391	0.9402	0.9391	0.9386
Random Forest	0.9670	0.9681	0.9670	0.9668
Proposed Extra Trees	0.9757	0.9763	0.9757	0.9755

Among the classical baselines evaluated under the same feature settings, the proposed Extra Trees model achieves the highest performance with 97.57% accuracy. The results indicate that performance gains are primarily driven by the quality of statistical feature engineering rather than model complexity.

1) *Class-wise Performance*: The class-wise performance of the model is summarized in Table V.

TABLE V
CLASS-WISE PERFORMANCE OF THE PROPOSED MODEL

Activity	Precision	Recall	F1-score	Support
Downstairs	0.96	0.97	0.96	391
Jogging	0.98	0.98	0.98	1298
Sitting	0.95	0.96	0.95	163
Standing	0.94	0.95	0.94	139
Upstairs	0.97	0.96	0.96	587
Walking	0.99	0.99	0.99	1400

The model achieves consistently high performance across all activities, as shown in Table V, with minor degradation for low-motion classes such as Sitting and Standing due to higher inter-class similarity.

2) *Confusion Matrix*: The classification behavior is further illustrated in Fig. 2.

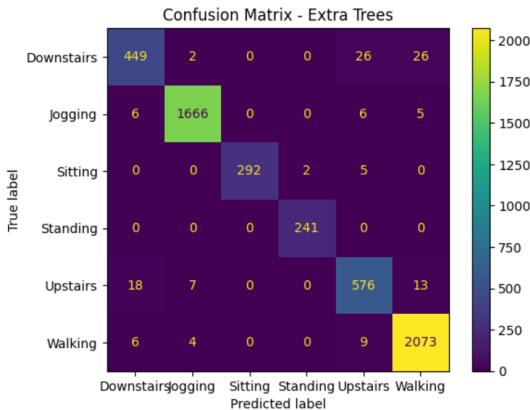


Fig. 2. Confusion matrix showing strong diagonal dominance and minimal confusion between classes.

The confusion matrix confirms strong classification performance, with most predictions concentrated on the diagonal. Errors mainly occur between similar activities such as Upstairs and Downstairs.

I. Visualization and Feature Analysis

1) *ROC Curve*: The model separability is visualized in Fig. 3.

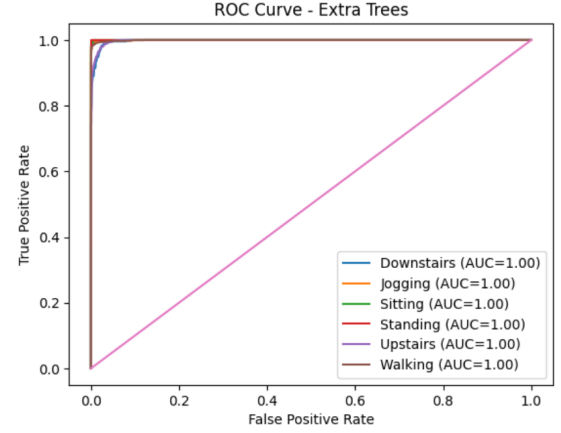


Fig. 3. ROC curves showing near-perfect separability across all classes.

The ROC analysis shows near-perfect separability, with AUC values close to 1.0, indicating strong discriminative capability.

2) *PCA Visualization*: The feature distribution in reduced space is shown in Fig. 4.

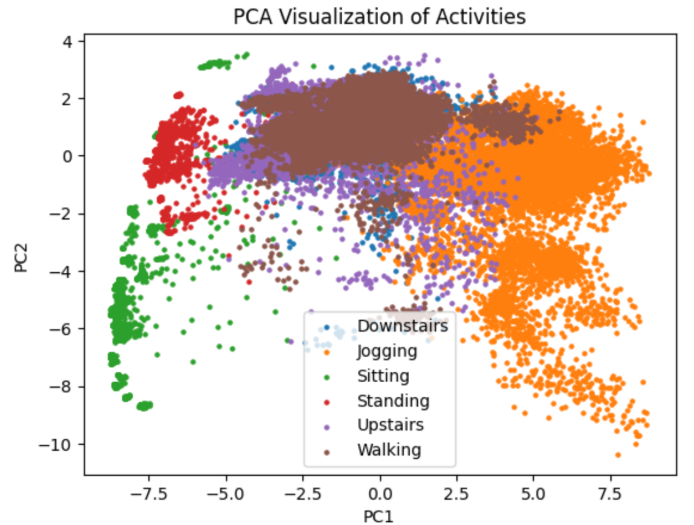


Fig. 4. PCA projection showing partial class separability in reduced space.

PCA shows partial class separability with overlap between similar activities, suggesting that higher-dimensional features are more discriminative.

3) *t-SNE Visualization*: Non-linear feature clustering is presented in Fig. 5.

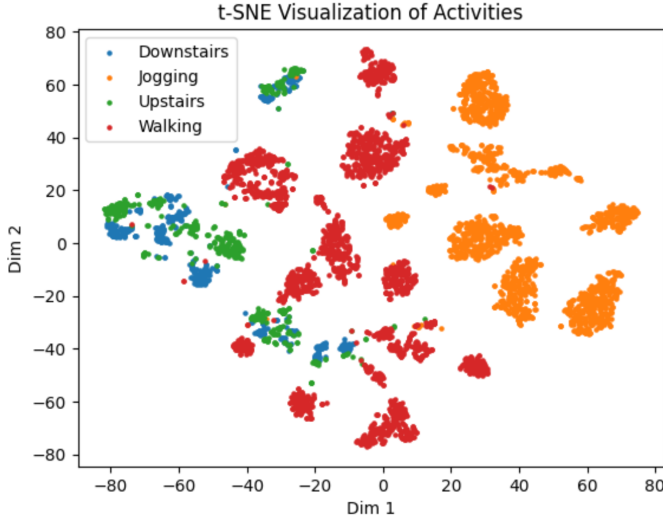


Fig. 5. t-SNE visualization showing well-formed clusters of activity classes.

t-SNE shows clearer non-linear clustering, indicating that the extracted features preserve meaningful activity-related structure.

4) *Feature Importance Analysis*: The contribution of features is highlighted in Fig. 6.

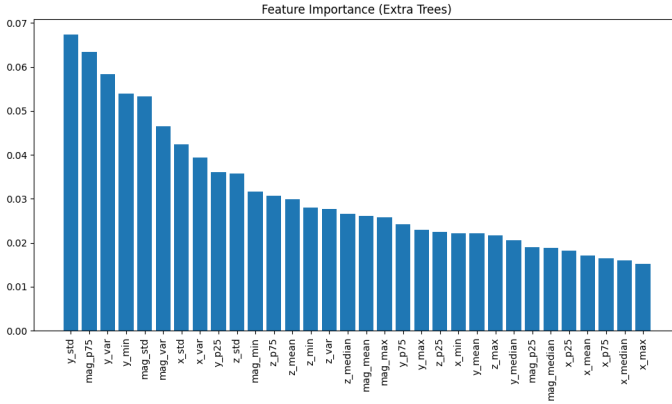


Fig. 6. Feature importance showing dominance of magnitude and variance-based features.

Feature importance analysis shows that magnitude-based and variance-related features contribute most to the classification decision, while other features have lower impact.

J. Evaluation Protocol and Fairness

A stratified split is used to preserve class distribution, and all in-study models are evaluated under the same feature settings. Comparisons with previously reported deep learning methods are used only as contextual references, since prior studies may differ in preprocessing, window sizes, train-test splits, hyperparameters, and evaluation protocols. Therefore,

the fairest comparisons in this work are those conducted under the same setup, including Logistic Regression, SVM, Random Forest, XGBoost, LightGBM, and Extra Trees.

1) *Ablation Study*: To analyze the contribution of key feature components, an ablation study is conducted, as shown in Table VI.

TABLE VI
ABLATION STUDY ON FEATURE COMPONENTS

Feature Configuration	Accuracy
Mean Only	0.9205
Mean + Std	0.9597
Without Variance Feature	0.9678
Without Std Feature	0.9664
Without Magnitude Feature	0.9731
Full Feature Set (Proposed)	0.9757

The results show that performance is not driven by a single feature type, but by the complementary effect of statistical dispersion features and magnitude-based representation.

2) *Comparison with State-of-the-Art*: To provide broader context, Table VII reports representative WISDM results from prior HAR studies. These results are not direct fair comparisons due to differences in preprocessing, segmentation, and evaluation protocols.

TABLE VII
CONTEXTUAL COMPARISON WITH REPORTED HAR METHODS ON WISDM

Study	Method	Accuracy (%)	Dataset
CNN [3]	CNN	92.59	WISDM
LSTM [3]	LSTM	97.40	WISDM
BiLSTM [2]	BiLSTM	97.86	WISDM
BiGRU [2]	BiGRU	97.96	WISDM
CNN-LSTM [7]	CNN-LSTM	95.29	WISDM
ResNet-BiGRU-SE [2]	Attention Model	98.78	WISDM
Proposed	Extra Trees	97.57	WISDM

The proposed model achieves competitive reported performance using a simpler feature-based pipeline, but direct superiority over deep learning models is not claimed.

V. COMPARISON WITH GRADIENT BOOSTING METHODS

To further evaluate the proposed framework, Extra Trees was compared with XGBoost and LightGBM, as shown in Table VIII.

TABLE VIII
COMPARISON WITH GRADIENT BOOSTING MODELS

Model	Accuracy
Extra Trees (Proposed)	0.9766
XGBoost	0.9764
LightGBM	0.9788

LightGBM achieves the highest accuracy, while Extra Trees remains closely competitive. Therefore, Extra Trees is not claimed to be universally superior, but is considered a balanced option due to its simplicity, parallelizability, limited tuning requirements, and compatibility with tree-based interpretability tools.

A. Computational Efficiency Comparison

Table IX reports training and inference times.

TABLE IX
COMPUTATIONAL EFFICIENCY COMPARISON

Model	Train (s)	Infer (ms)
Extra Trees (Prop.)	24.55	0.296
XGBoost	35.78	0.038
LightGBM	13.95	0.133

Extra Trees provides a balanced trade-off between accuracy, simplicity, and interpretability, while LightGBM achieves faster training and XGBoost achieves lower latency. Real-device deployment and energy validation remain necessary for practical edge use.

VI. EXPLAINABILITY ANALYSIS USING SHAP

To enhance interpretability, SHAP (SHapley Additive exPlanations) is used to quantify feature contributions in the Extra Trees classifier, as shown in Fig. 7.

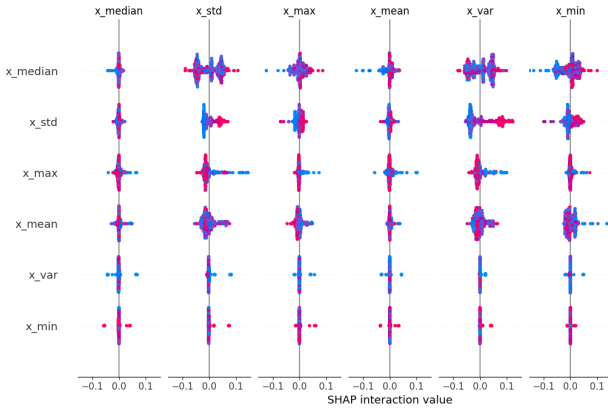


Fig. 7. SHAP interaction plot showing feature contribution to model predictions.

SHAP analysis shows that variance and standard deviation are the most influential features, indicating that the model relies mainly on dispersion-based patterns rather than absolute signal values.

A. Discussion

The proposed Extra Trees framework achieves 97.57% accuracy on WISDM and outperforms the classical baselines evaluated under the same setup. However, LightGBM achieves the highest accuracy among the gradient boosting models, so the proposed approach should be viewed as a balanced lightweight solution rather than a universally superior model. Ablation, feature importance, and SHAP analyses show that the main gains come from magnitude-based and dispersion-aware statistical features, while UCI HAR evaluation indicates reasonable cross-dataset generalization. Real-device validation remains necessary for confirming practical deployment.

VII. LIMITATIONS

Although the proposed framework achieves competitive performance with low computational cost, several limitations should be acknowledged. First, the contribution of magnitude-enhanced statistical features is incremental, since statistical feature engineering and ensemble classifiers are already widely studied in HAR. The main value of this work lies in the systematic combination of ablation analysis, cross-dataset evaluation, explainability, and efficiency assessment. Second, comparisons with deep learning methods are based on reported results and may not be fully direct due to differences in pre-processing, training settings, window sizes, dataset splits, and evaluation protocols. Third, although the framework shows low software-level inference latency and compact model size, real wearable or edge-device deployment and energy-consumption measurements have not yet been conducted.

VIII. CONCLUSION AND FUTURE WORK

This paper presented a lightweight HAR framework based on magnitude-enhanced statistical features and an Extra Trees classifier. The proposed approach achieved 97.57% accuracy on WISDM, outperforming the classical baselines evaluated under the same setup and remaining competitive with XGBoost and LightGBM. Overall, the results show that the framework provides a practical balance between accuracy, simplicity, efficiency, and interpretability.

Future work will focus on real wearable or edge-device deployment, energy-consumption measurements, and hybrid lightweight models that combine statistical features with compact temporal learning components.

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